

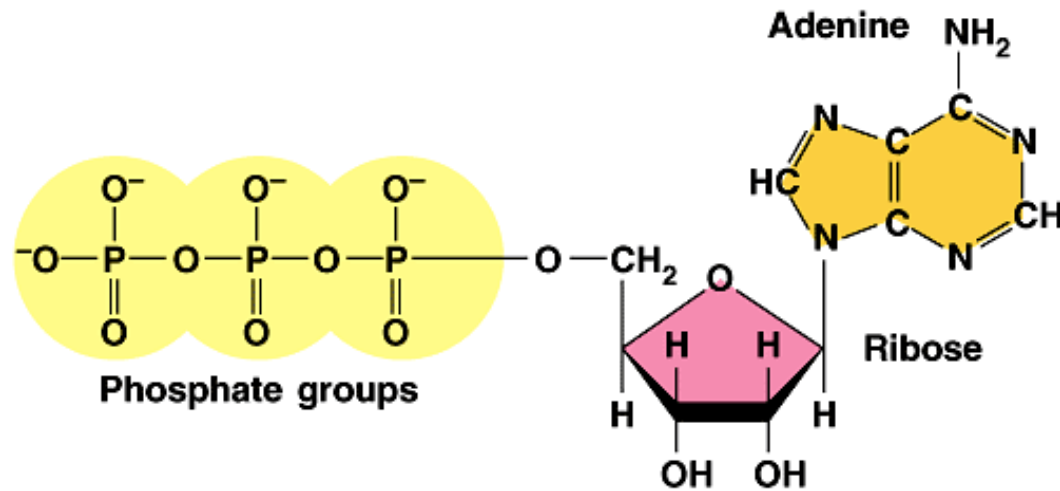
Energy of Life

Lecture 13-15

Metabolism

The sum of the chemical processes that occur in living organisms, resulting in growth, production of energy, elimination of waste material, etc.

- **Anabolism**- build up of complex molecules
- **Catabolism**- break down of complex molecules



Cellular Respiration

Harvesting Chemical Energy



Harvesting stored energy

- Energy is stored in organic molecules (e.g. **carbohydrates, fats, proteins**)
- **Heterotrophs** eat these organic molecules → **Food**
 - digest organic molecules to get...
 - **raw materials** for synthesis
 - **fuels** for energy
 - controlled release of energy
 - “burning” fuels in a series of **step-by-step enzyme-controlled reactions**



ATP generation by different processes

1. Substrate-Level Phosphorylation: Direct transfer of a phosphate group from a high-energy substrate to ADP, forming ATP.

•**Examples:**

- **Glycolysis:** generating 2 ATP per glucose molecule.
- **Citric Acid Cycle (Krebs Cycle):** 1 ATP (or GTP) per turn.

2. Oxidative Phosphorylation: Involves the electron transport chain (ETC) and chemiosmosis, similar to eukaryotic cells but occurring in the plasma membrane in bacteria. 28-34 ATP per glucose molecule, depending on the efficiency of the ETC. Electrons derived from NADH and FADH₂ pump protons (H⁺) across the membrane, creating a proton gradient.

3. Photophosphorylation (in Photosynthetic Bacteria): Some bacteria, such as purple and green sulfur bacteria, generate ATP using light energy, depending on the light intensity and specific bacterial species.

•**Steps:**

- Light energy excites electrons in photosynthetic pigments, which are transferred through an electron transport chain.
- This process also creates a proton gradient, similar to oxidative phosphorylation.
- Protons flow back through ATP synthase to generate ATP.

ATP generation by different processes.....2

4. Anaerobic Respiration: Similar to aerobic respiration, but uses alternative electron acceptors instead of oxygen. ATP yield can vary depending on the electron acceptor used.

• **Acceptors:** Nitrate (NO_3^-), sulfate (SO_4^{2-}), or carbon dioxide (CO_2) can serve as terminal electron acceptors.

5. Fermentation: Anaerobic process that allows ATP production in the absence of oxygen. Allows survival in anaerobic conditions.

- **Lactic Acid Fermentation:** Some bacteria, like *Lactobacillus*, convert glucose into lactic acid, yielding 2 ATP per glucose.
- **Alcoholic Fermentation:** Certain bacteria (and yeasts) convert glucose into ethanol and CO_2 , also producing 2 ATP per glucose

Overview of cellular respiration

- 4 metabolic stages

1. Glycolysis

- Anaerobic respiration

- respiration without O_2
 - in cytosol

- Aerobic respiration

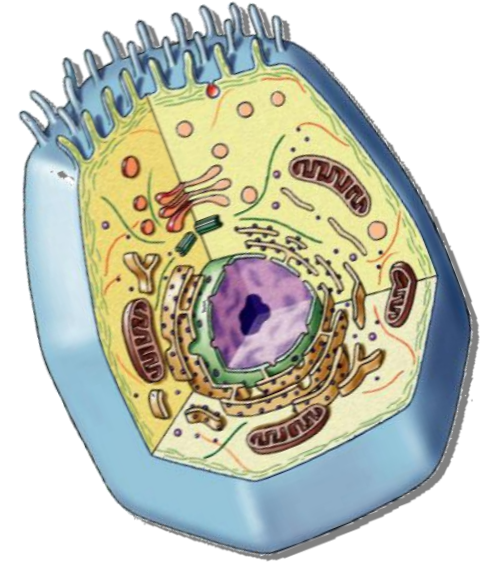
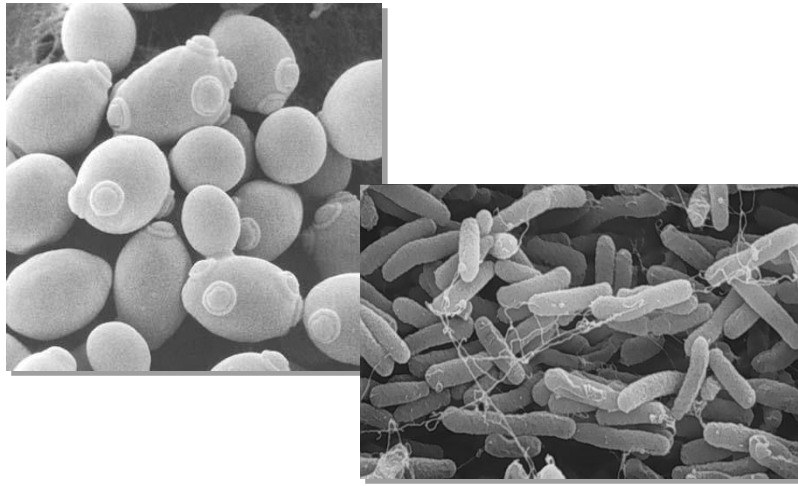
- respiration using O_2
 - in mitochondria

2. Pyruvate oxidation

3. Krebs cycle

4. Electron transport chain





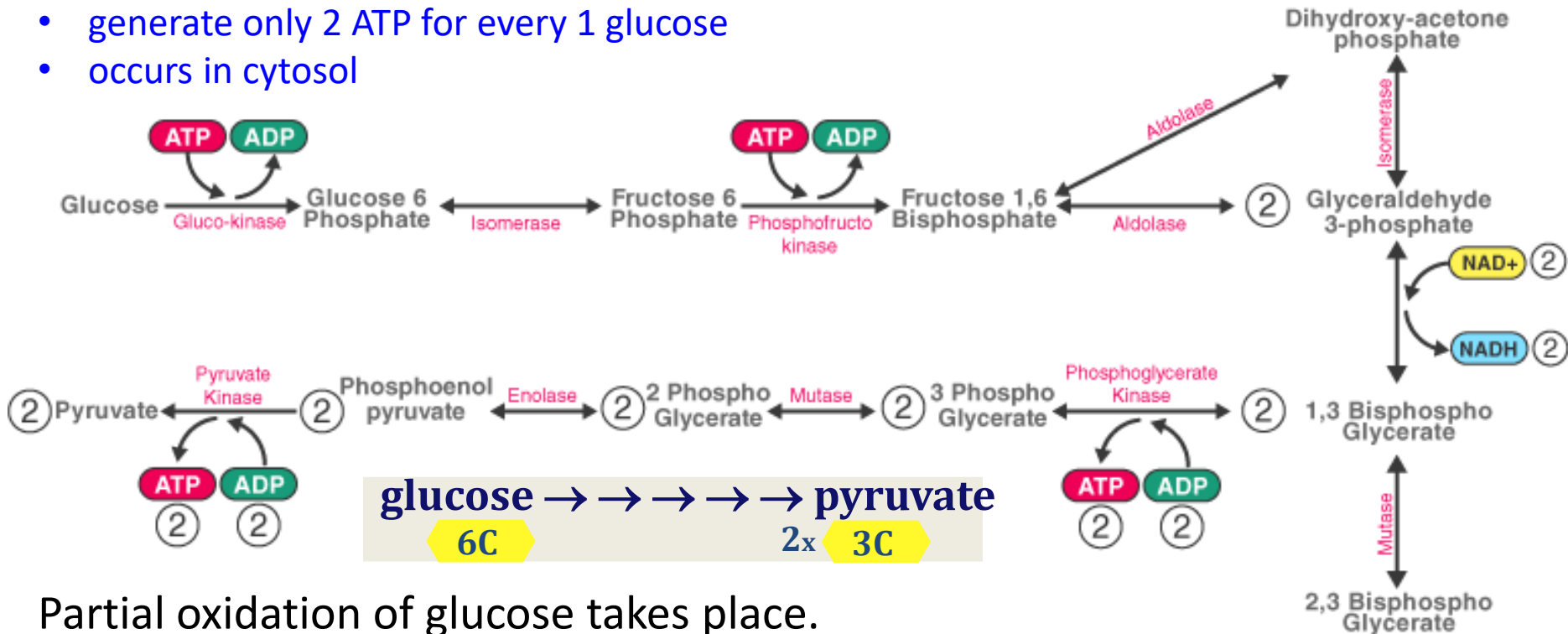
Cellular Respiration

Stage 1: Glycolysis



Glycolysis

- generate only 2 ATP for every 1 glucose
- occurs in cytosol



Partial oxidation of glucose takes place.

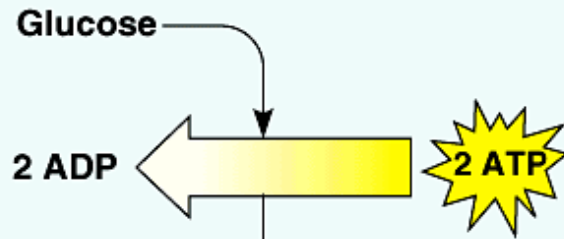


It occurs in two phases:

- **Preparatory Phase or Energy-requiring Phase:** The first 5 steps of the EMP pathway are known as the investment phase. Energy is consumed in this process to produce two molecules of triose sugar phosphates.
- **Pay-off Phase:** The second half of glycolysis produces ATP and NADH and pyruvate.

Glycolysis summary

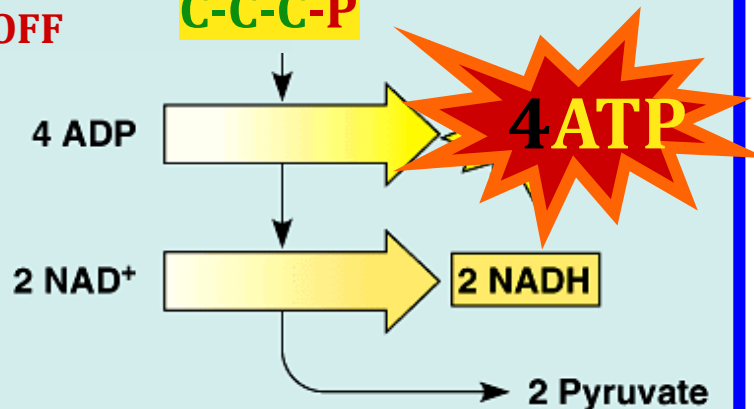
ENERGY INVESTMENT



Endergonic

invest some ATP

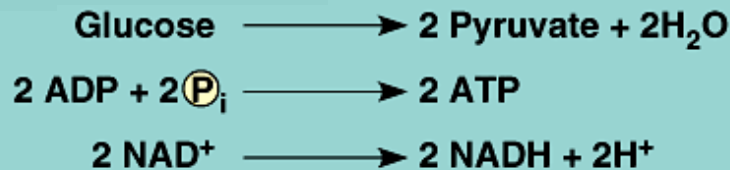
ENERGY PAYOFF



Exergonic

harvest a little
ATP & a little NADH

NET YIELD

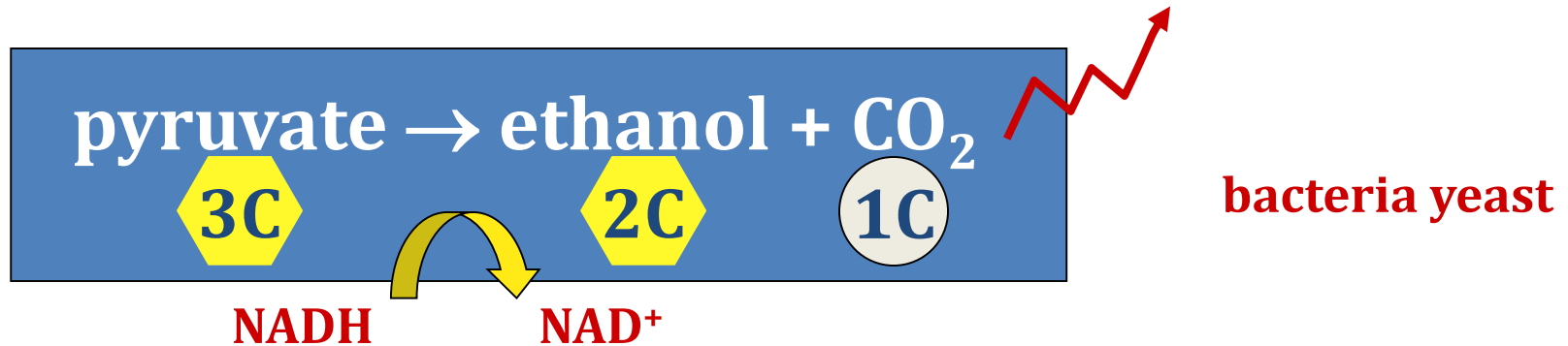


Yield

2 ATP

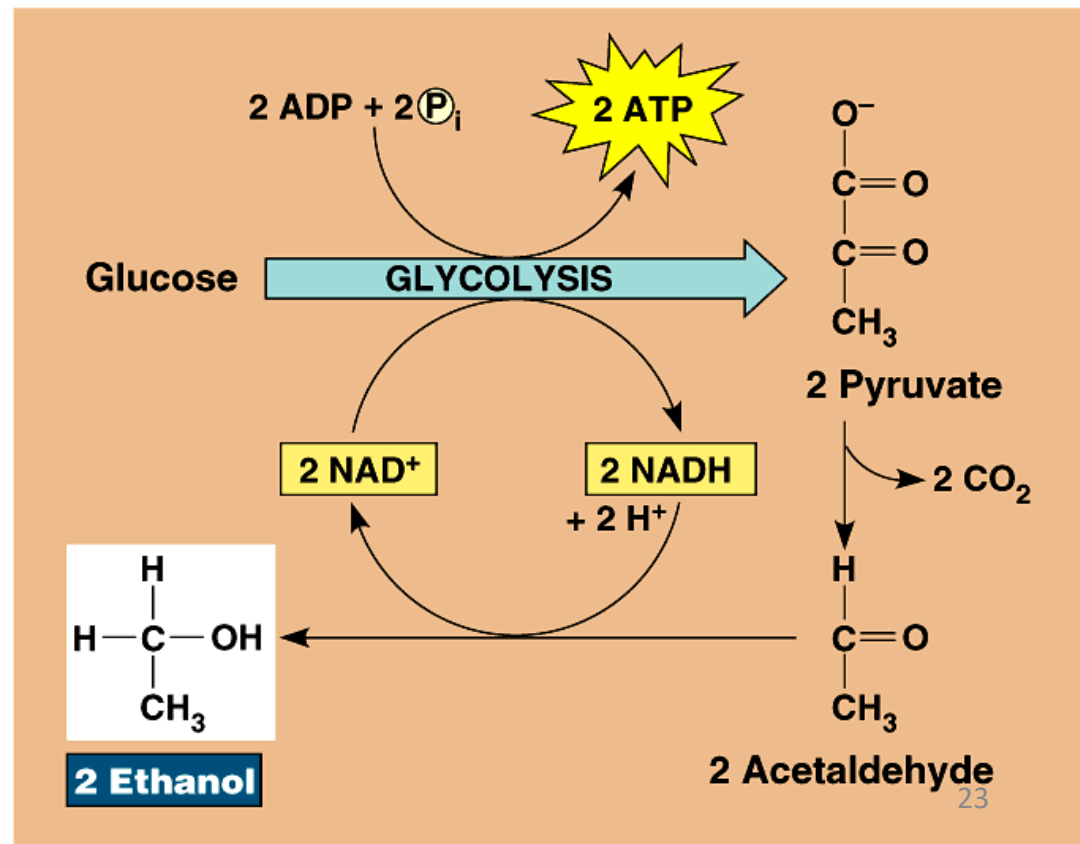
2 NADH

Alcohol Fermentation



Dead end process

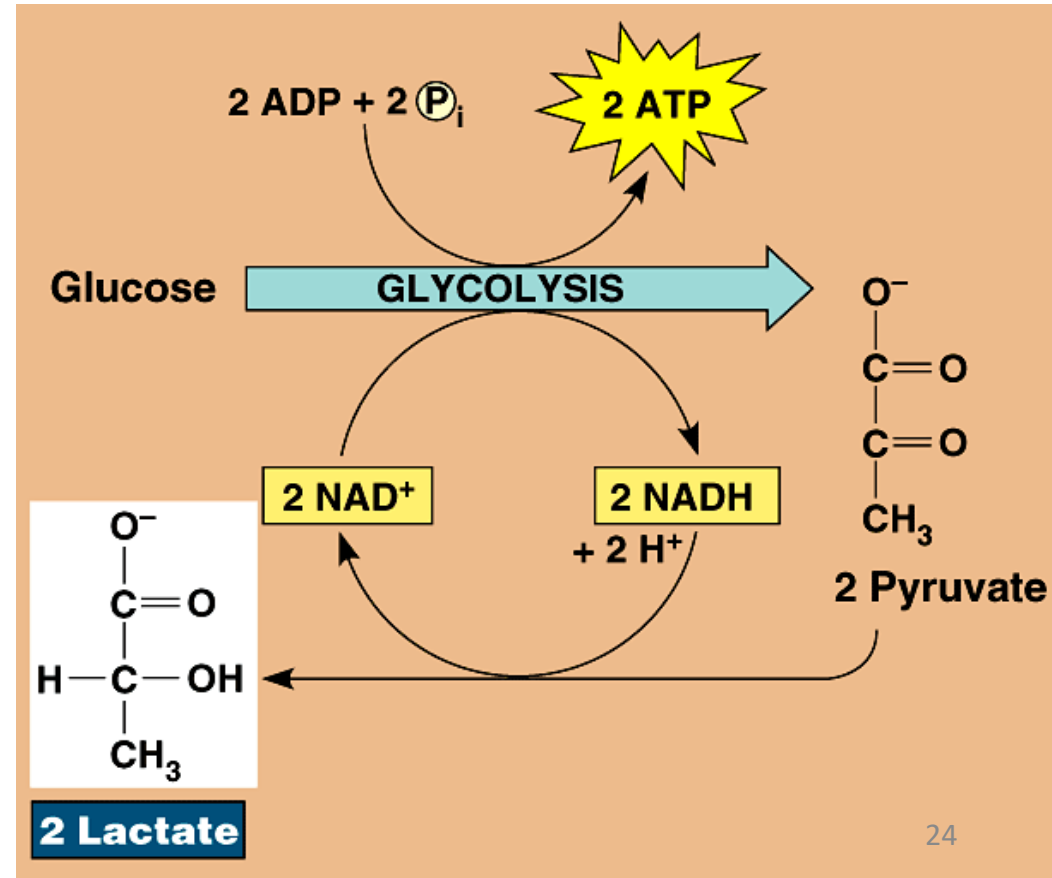
- at ~12% ethanol, kills yeast
- can't reverse the reaction



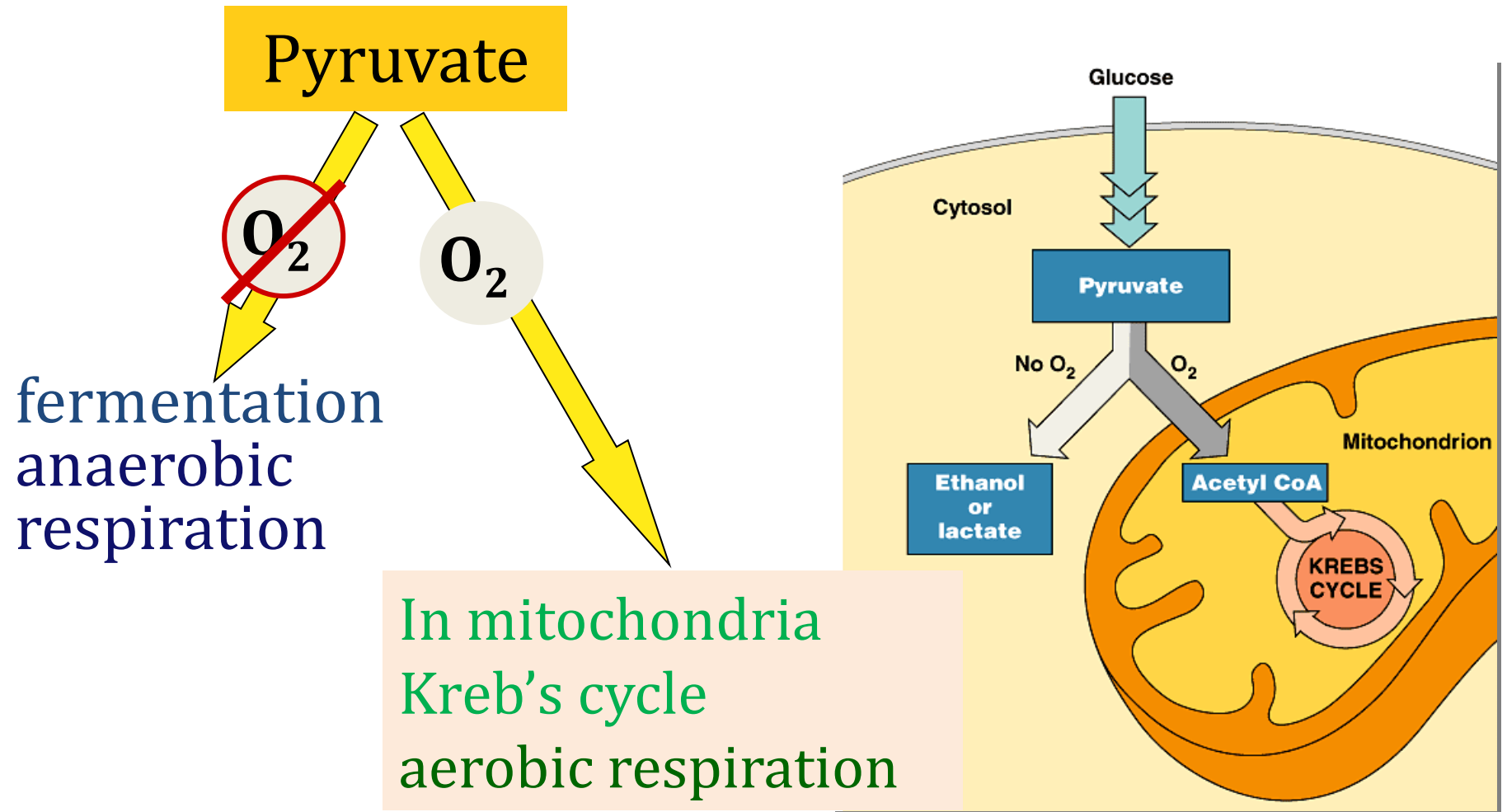
Lactic Acid Fermentation



- **Reversible process**
 - once O_2 is available, lactate is converted back to pyruvate by the liver



Pyruvate is a branching point



Cellular Respiration

Stage 2 & 3:

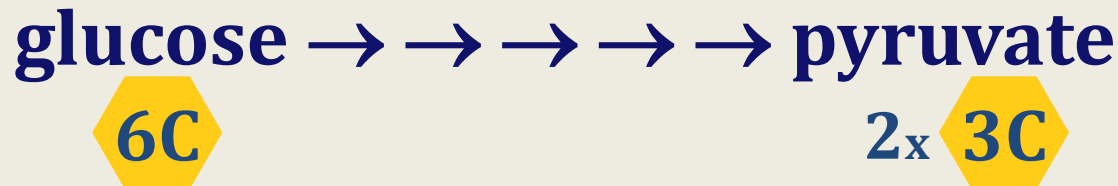
Oxidation of Pyruvate

Krebs Cycle

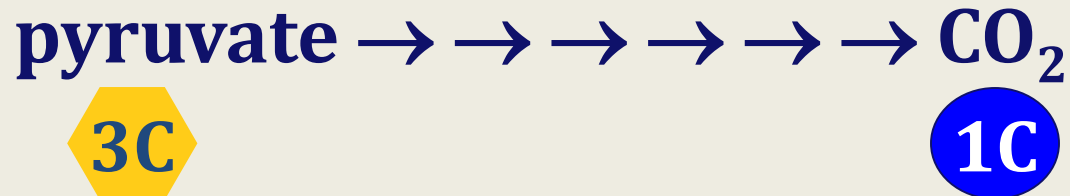


Glycolysis is only the start

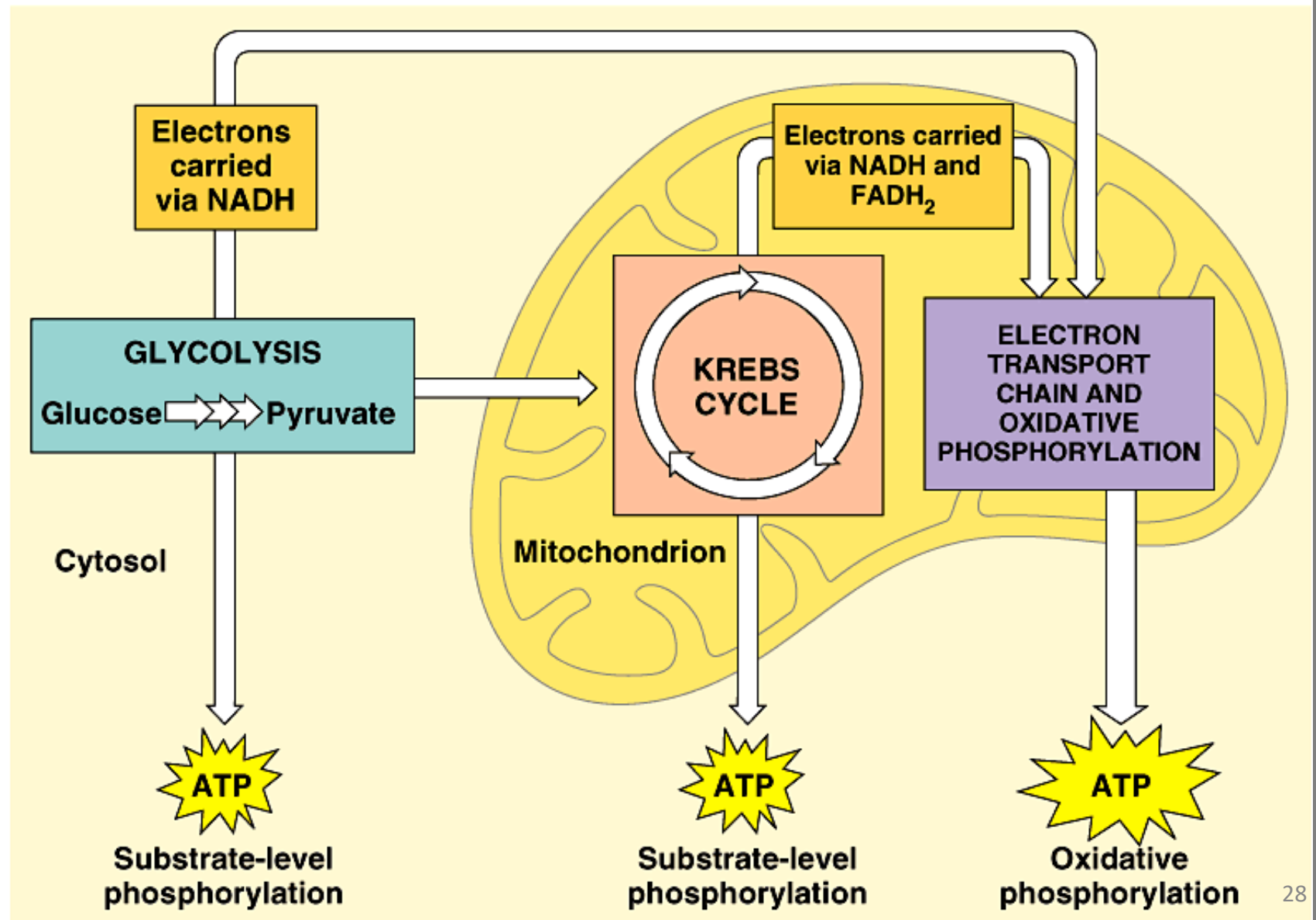
- **Glycolysis**



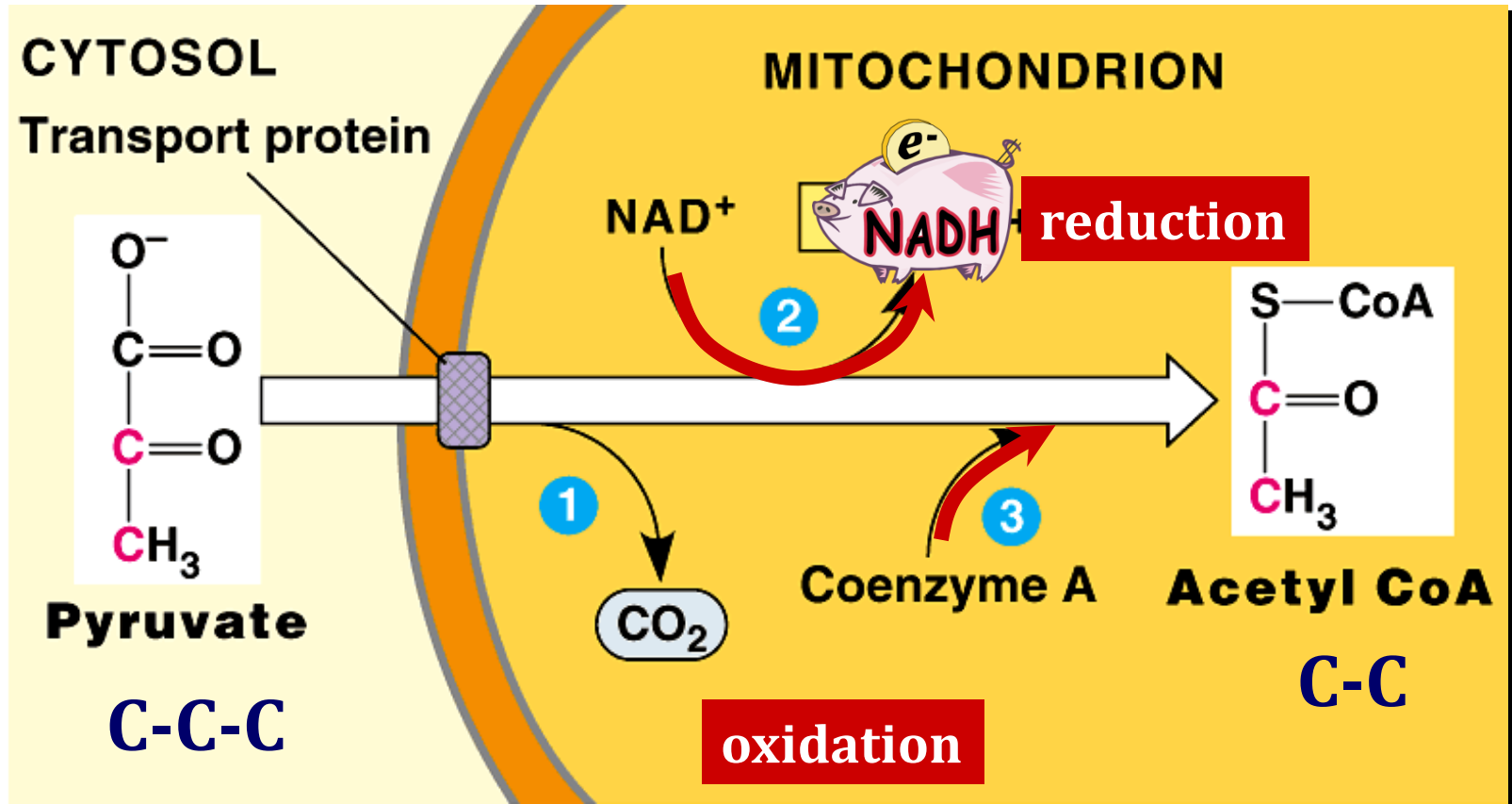
- **Pyruvate has more energy to yield**
 - if O_2 is available, pyruvate enters mitochondria
 - enzymes of Krebs cycle complete the full oxidation of sugar to CO_2



Cellular respiration



Pyruvate oxidized to Acetyl CoA



Krebs cycle

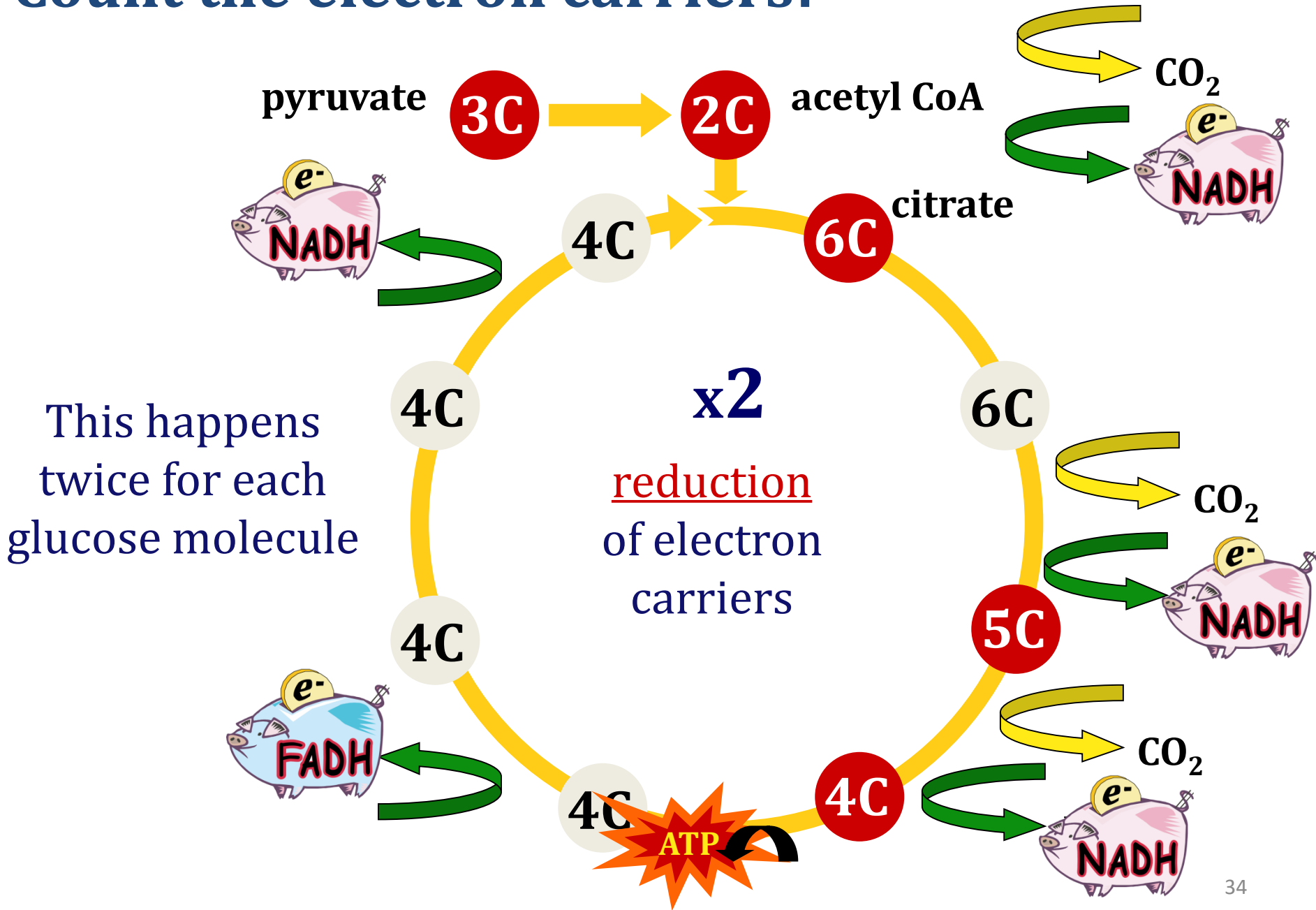
1937 | 1953



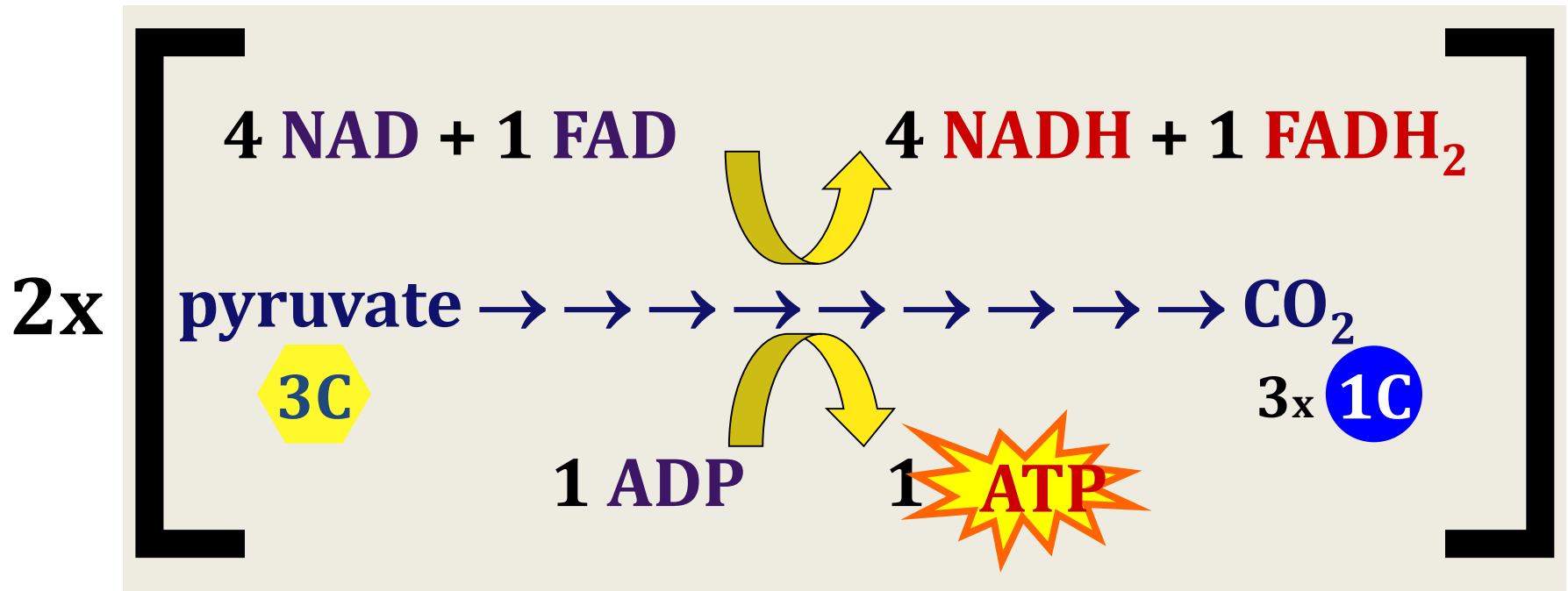
Hans Krebs
1900-1981

- Citric Acid Cycle
 - in mitochondrial matrix
 - 8 step pathway
 - each catalyzed by specific enzyme
 - step-wise catabolism of 6C citrate molecule
- Evolved later than glycolysis
 - does that make evolutionary sense?
 - bacteria → 3.5 billion years ago (**glycolysis**)
 - free O₂ → 2.7 billion years ago (**photosynthesis**)
 - eukaryotes → 1.5 billion years ago (**aerobic respiration = organelles → mitochondria**)

Count the electron carriers!



Energy accounting of Krebs cycle

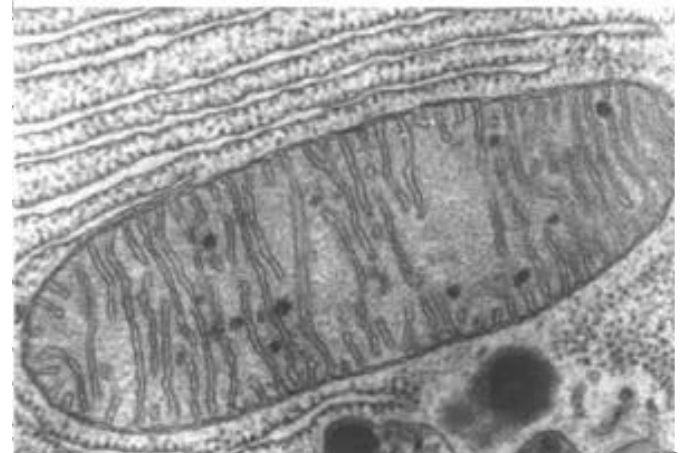
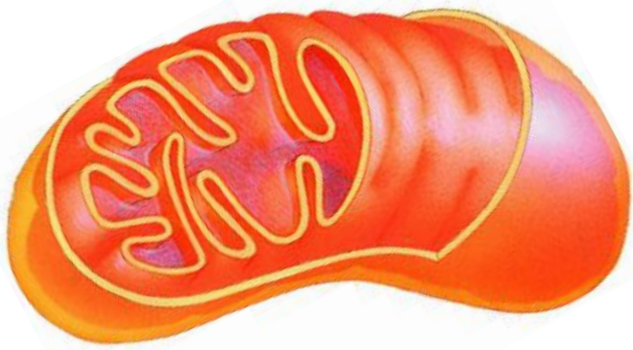


$$\begin{aligned}\text{Net gain} &= 2 \text{ ATP} \\ &= 2 + 6 \text{ NADH} + 2 \text{ FADH}_2\end{aligned}$$

Cellular Respiration

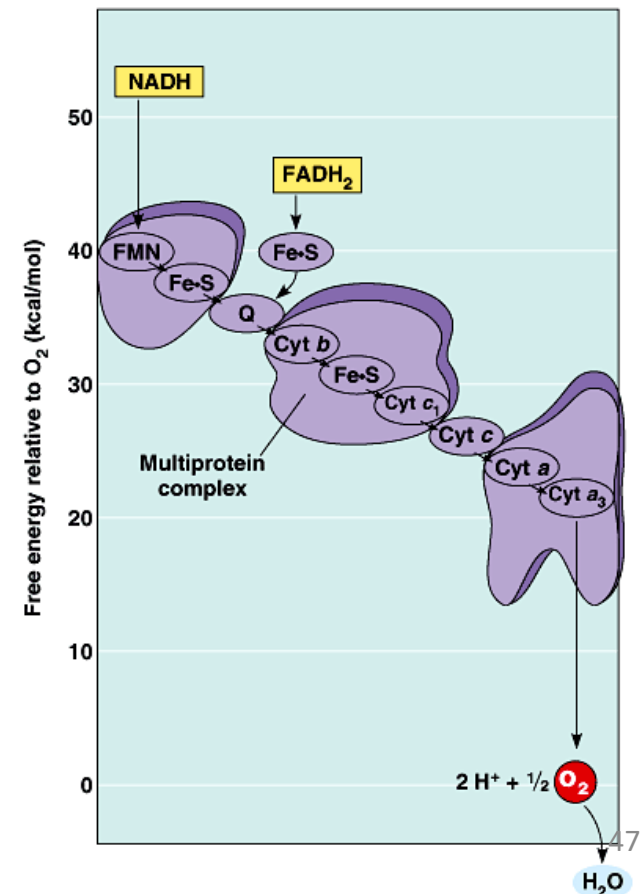
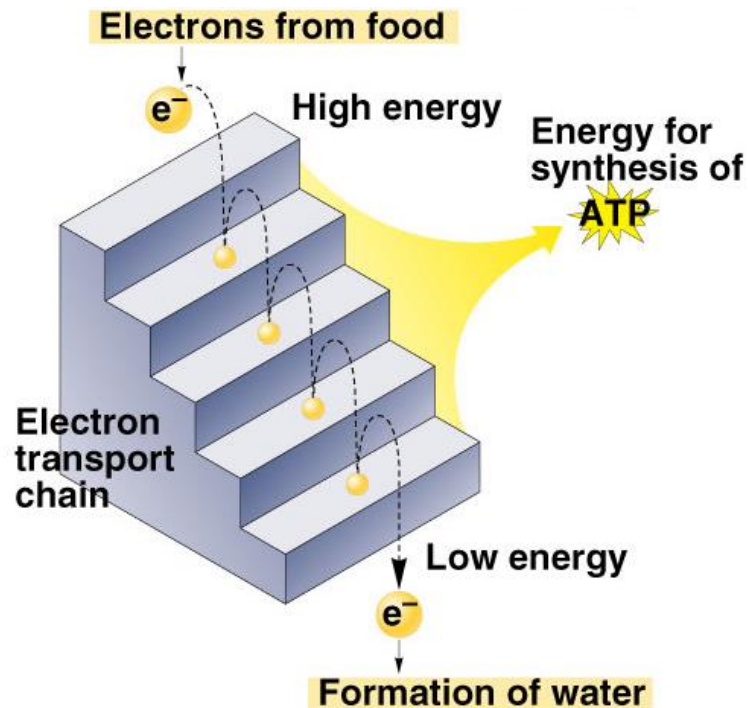
Stage 4:

Electron Transport Chain

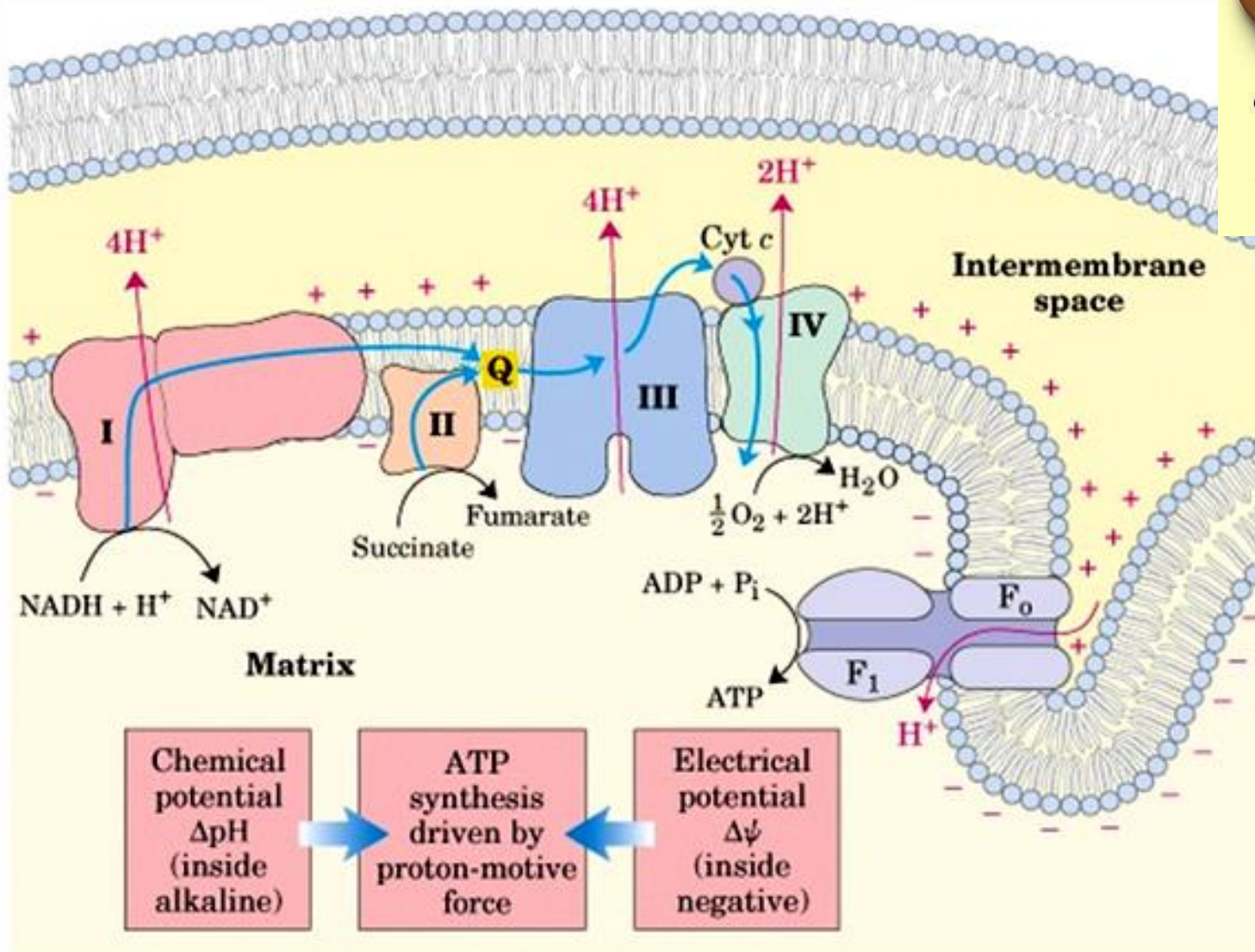
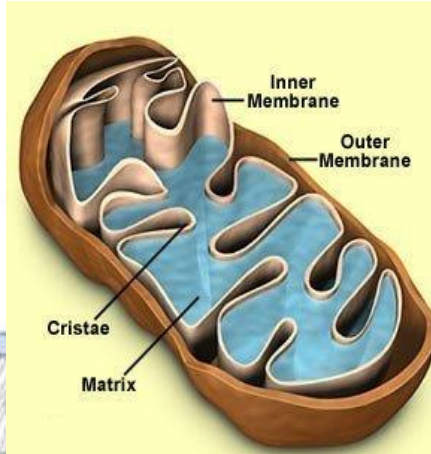


Electrons flow down hill

- Electrons move in steps from carrier to carrier **downhill to O_2**
 - each carrier is more electronegative
 - controlled oxidation
 - controlled release of energy



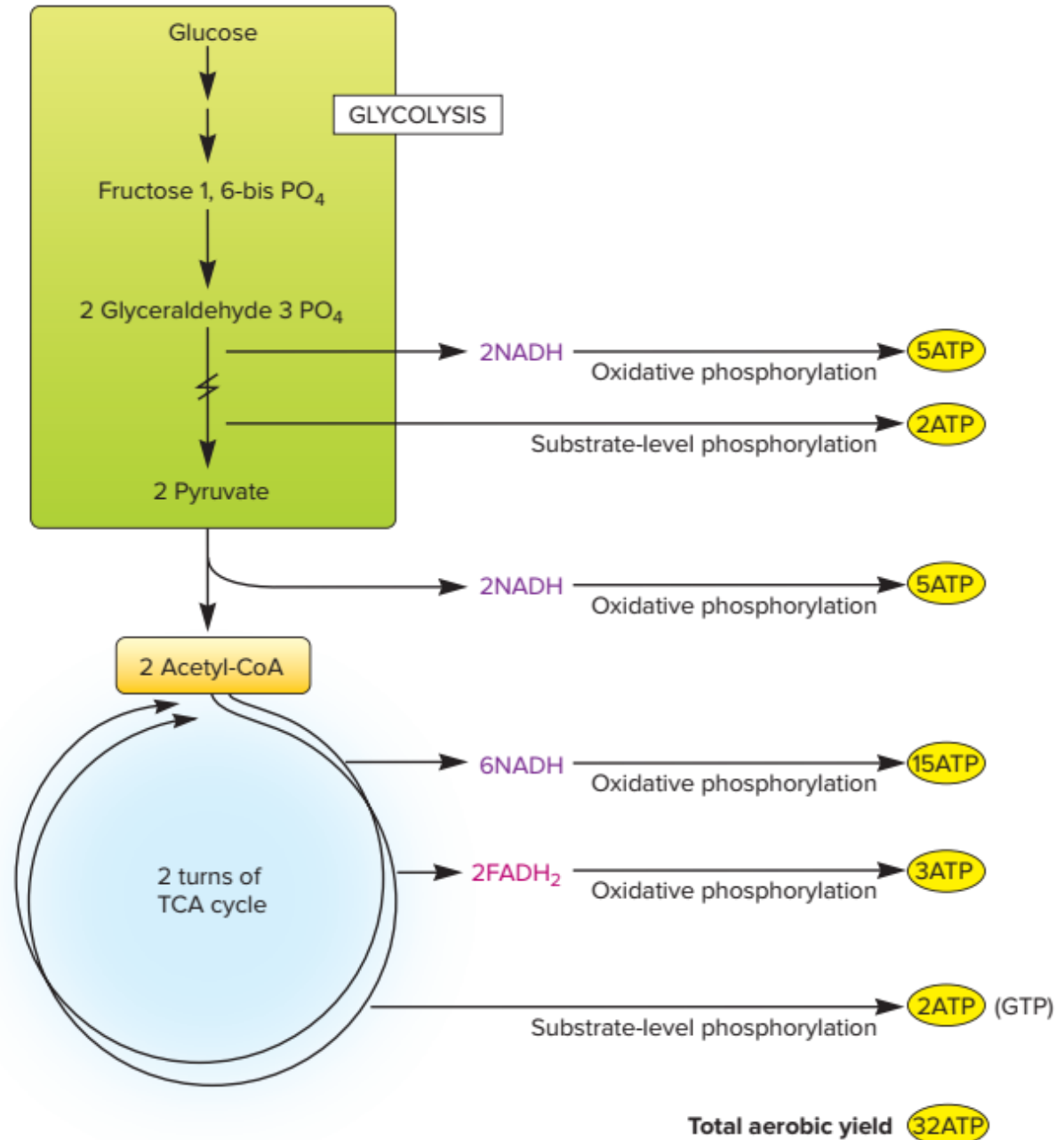
ETC of Respiration

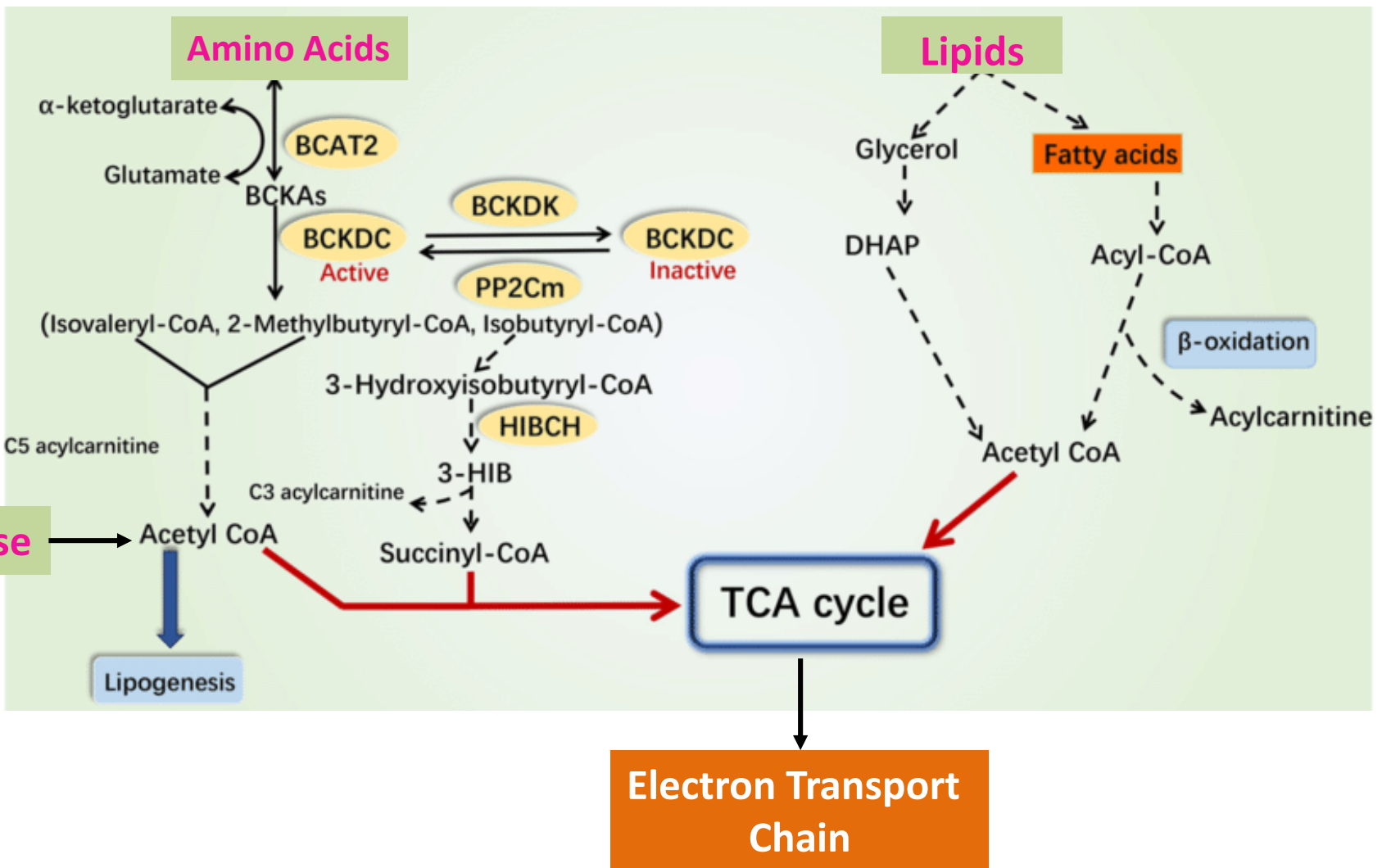


oxidative phosphorylation

Maximum Theoretic ATP Yield from Aerobic Respiration

- The best estimates of the number of protons moved across the membrane due to **NADH** oxidation is **10**; **six** are thought to be transferred **when FADH_2 oxidation** initiates electron flow to oxygen.
- It is currently thought that **four protons are consumed during ATP synthesis**; **three** are used by ATP synthase and **one** is used for transport of ATP, ADP, and P_i .
- Thus the P/O ratio is estimated to be **2.5 for NADH** and **1.5 for FADH_2** .





Enzymes

- A protein catalyst
- Enzymes are important proteins found in living things. An **enzyme** is a protein that changes the rate of a chemical reaction.

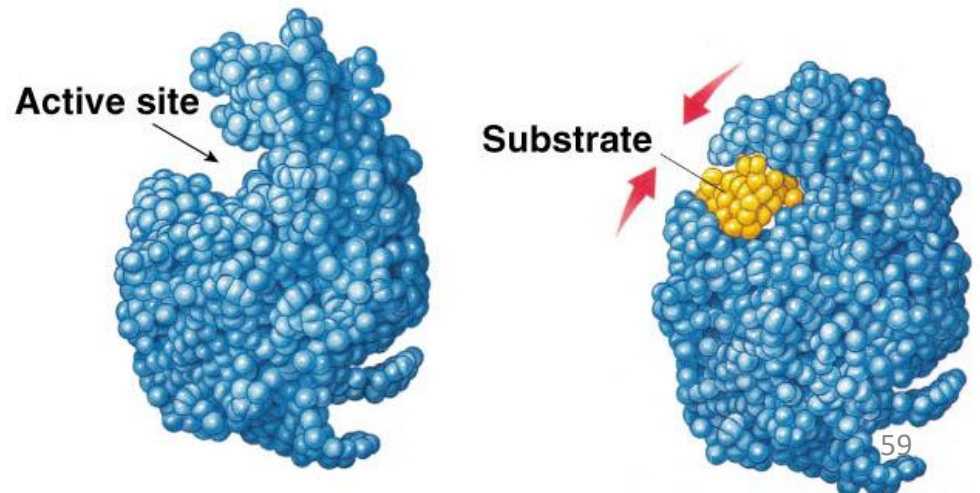
They speed up metabolic reactions.

- Each enzyme is the specific helper to a specific reaction
 - each enzyme needs to be the right shape for the job
 - enzymes are named for the reaction they help
 - sucrase breaks down sucrose
 - proteases breakdown proteins
 - lipases breakdown lipids
 - DNA polymerase builds DNA

Enzyme

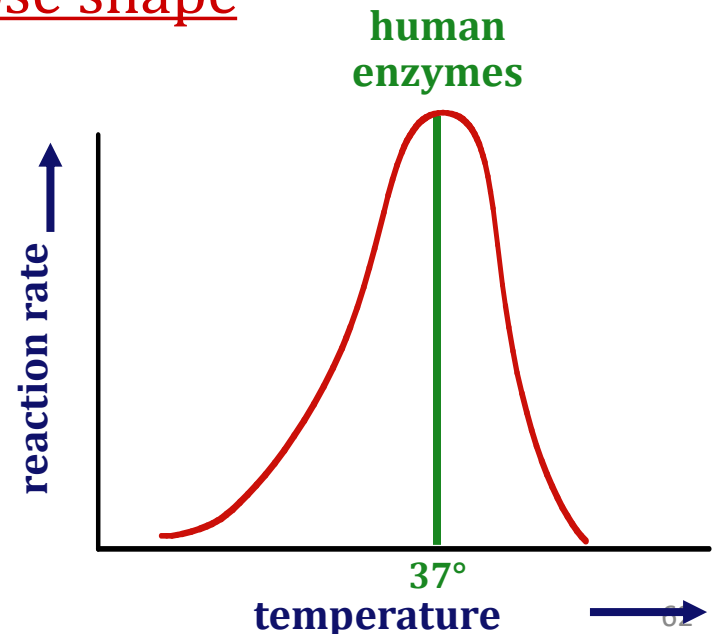
Enzymes are not changed by the reaction

- Enzyme
 - helper protein molecule
- Substrate
 - molecule that enzymes work on
- Products
 - what the enzyme helps produce from the reaction
- Active site
 - part of enzyme that substrate molecule fits into



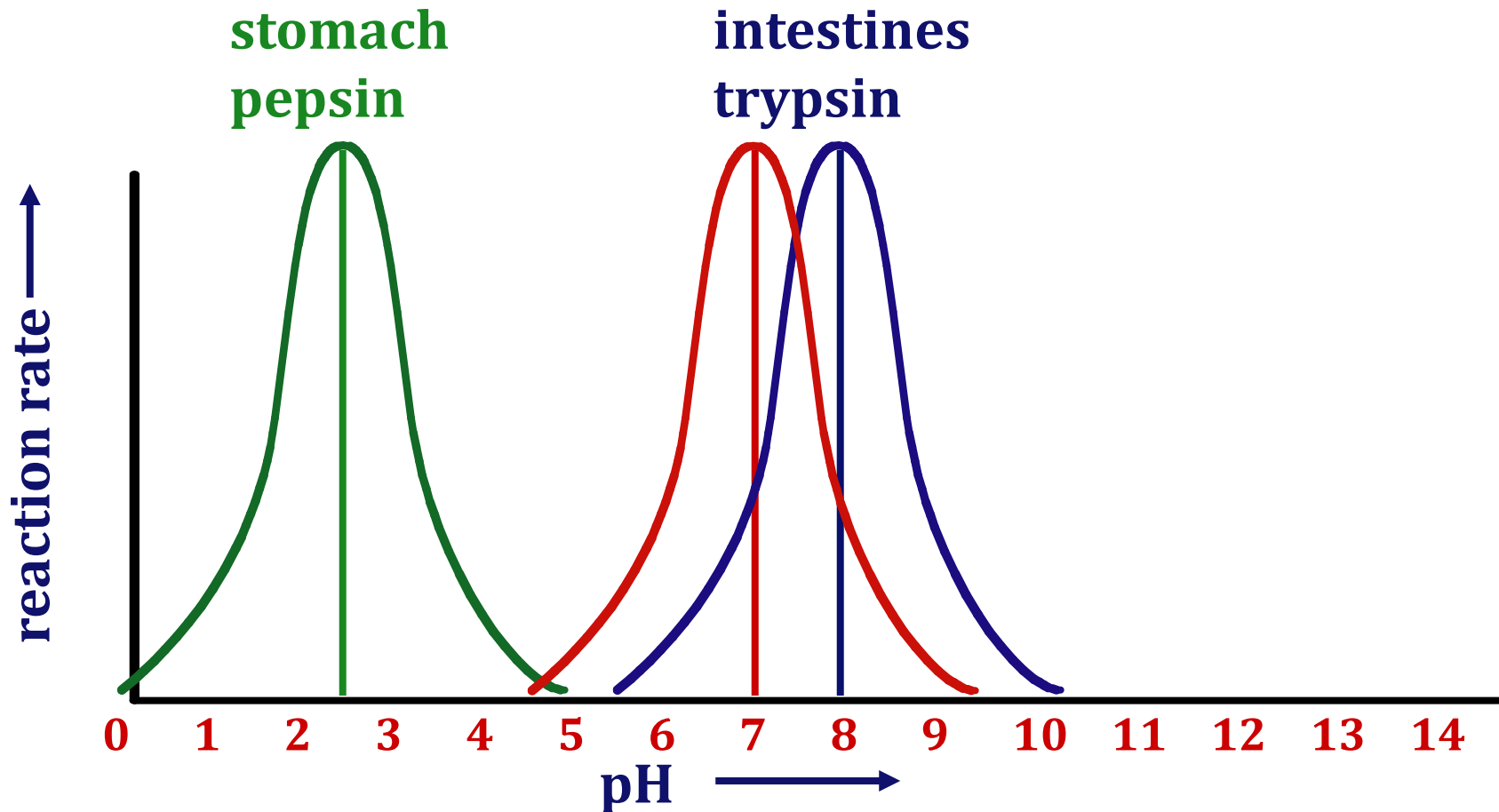
Temperature

- Effect on rates of enzyme activity
 - Optimum temperature
 - human enzymes
 - 35°- 40°C (body temp = 37°C)
 - Raise temperature (boiling)
 - denature protein = unfold = lose shape
 - Lower temperature T°
 - molecules move slower



pH

- Effect on rates of enzyme activity
 - changes in pH changes protein shape~ Denatures
 - most human enzymes = pH 6-8
 - depends on where in body
 - pepsin (stomach) = pH 3
 - trypsin (small intestines) = pH 8



Classification of Enzymes

Enzymes are protein. There are **6 types of reaction** they can catalyze-

1. **Oxidoreductases:** These catalyze oxidation and reduction reactions. **Alcohol dehydrogenase**
2. **Transferases:** These catalyze the transfer of a chemical group from one compound to another. **Kinases**
3. **Hydrolases:** These catalyze hydrolysis (splitting by use of water) reactions. Most digestive enzymes are hydrolases. **Lipase**
4. **Lyases:** These catalyze the breakdown of molecules by reactions that do not involve hydrolysis or Redox. **Fumarase**
5. **Isomerases:** These catalyze the transformation of one isomer into another (transfers a functional group from one position to another within a molecule). **Mutase**
6. **Ligases:** These catalyze the formation of bonds between compounds, often using the free energy made available from ATP hydrolysis. **DNA ligase**

To remember from the acronym: **OTHLIL**

Photosynthesis:

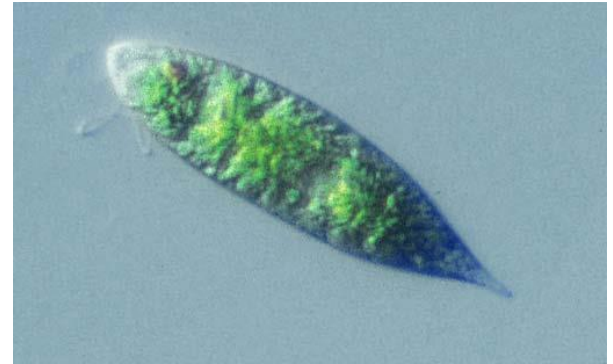
Life from **Light** and **Air**



Energy needs of life

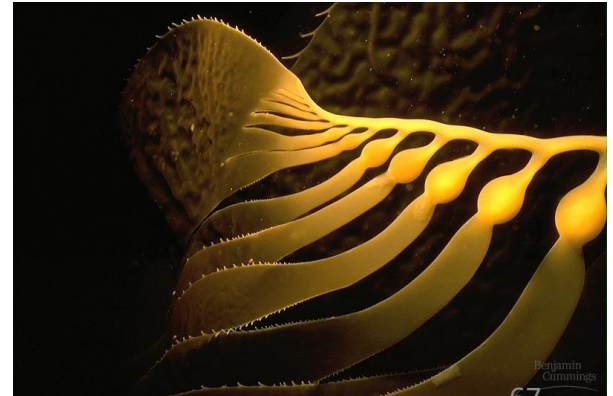
– Heterotrophs

- consumers
- animals
- fungi
- most bacteria



– Autotrophs

- producers
- plants
- photosynthetic bacteria (blue-green algae)



How are they connected?

Heterotrophs

making energy & organic molecules from ingesting organic molecules

glucose + oxygen → carbon + water + energy
dioxide



Autotrophs

making energy & organic molecules from light energy

carbon + water + energy → glucose + oxygen
dioxide



Plant structure

- Obtaining raw materials

- sunlight

- leaves = solar collectors

- CO₂

- stomates = gas exchange

- H₂O

- uptake from roots

- nutrients

- N, P, K, S, Mg, Fe...
 - uptake from roots

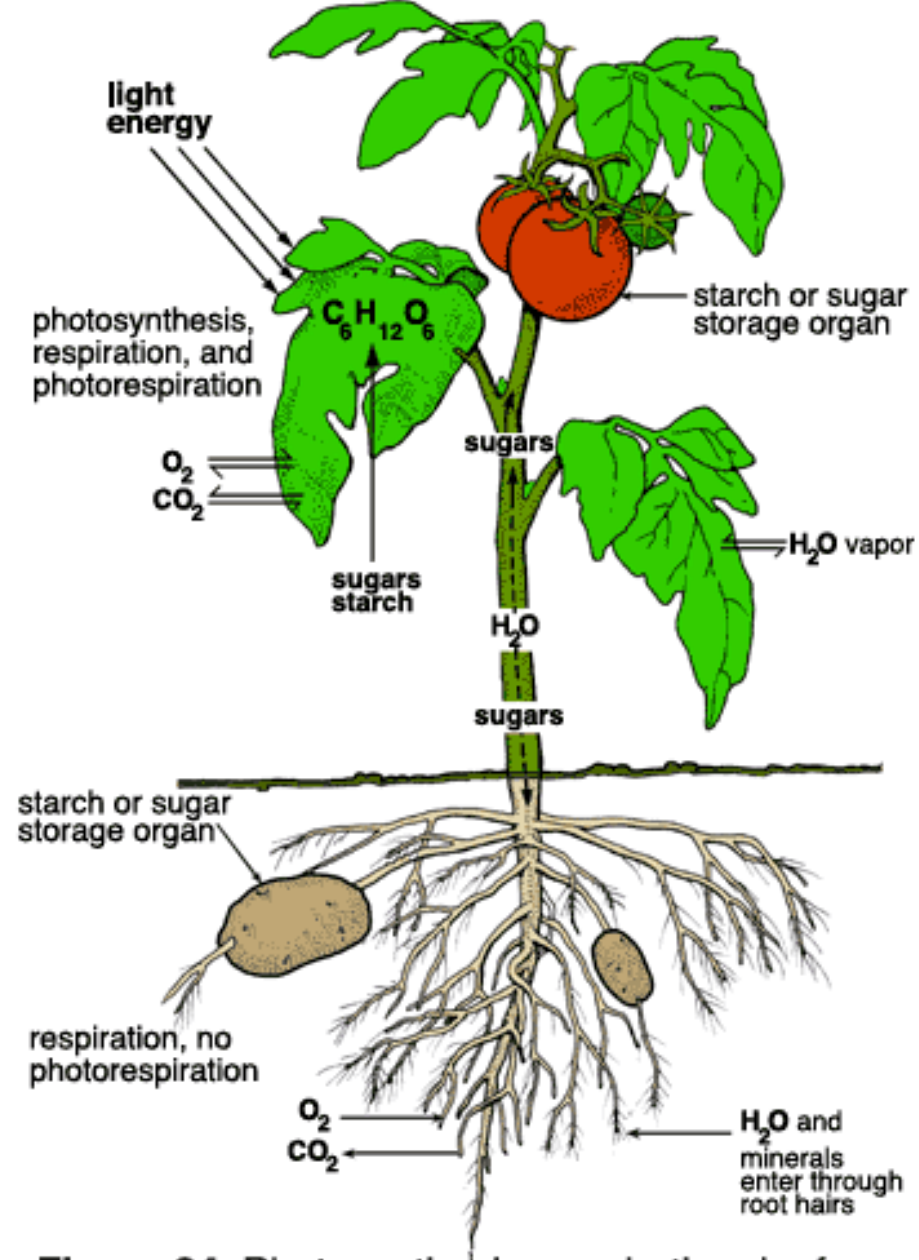
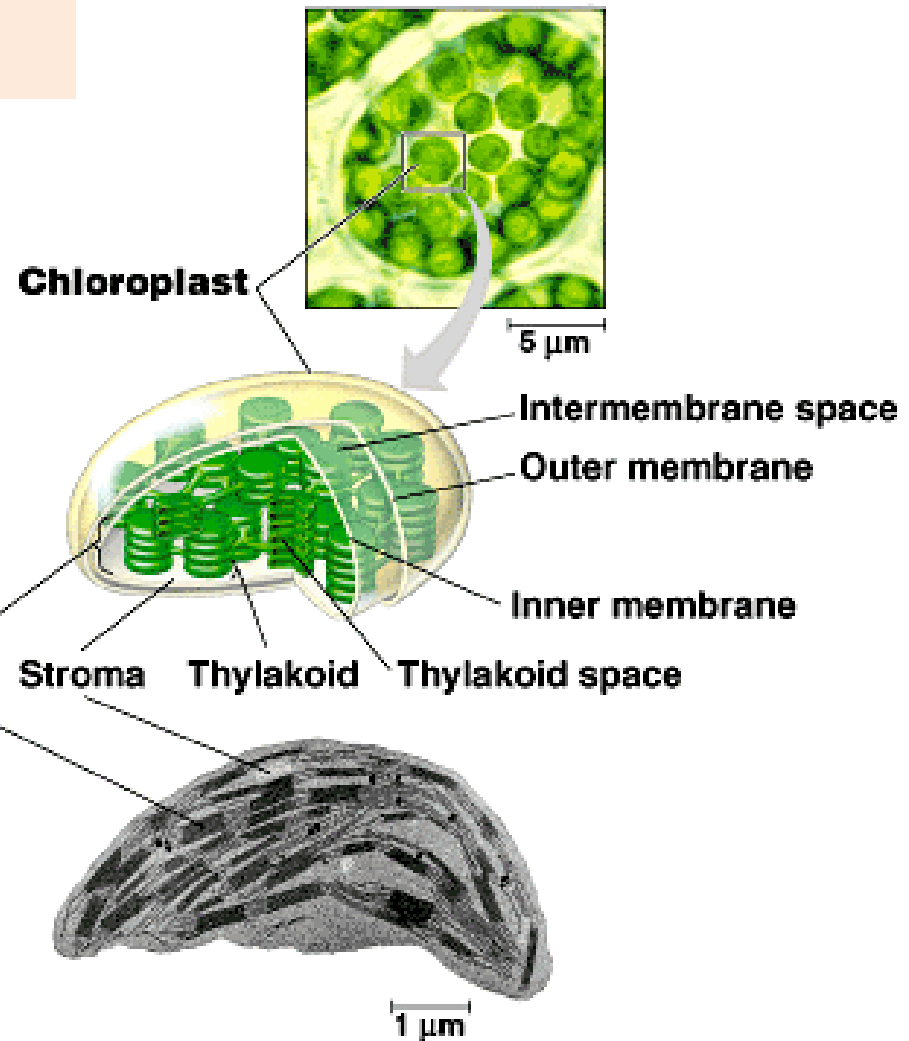


Figure 24. Photosynthesis, respiration, leaf water exchange, and translocation of sugar (photosynthate) in a plant.

Plant structure

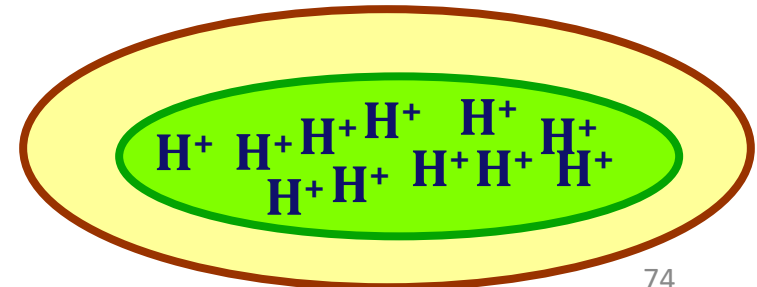
- **Chloroplasts**

- double membrane
- stroma
 - fluid-filled interior
- thylakoid sacs
- grana stacks

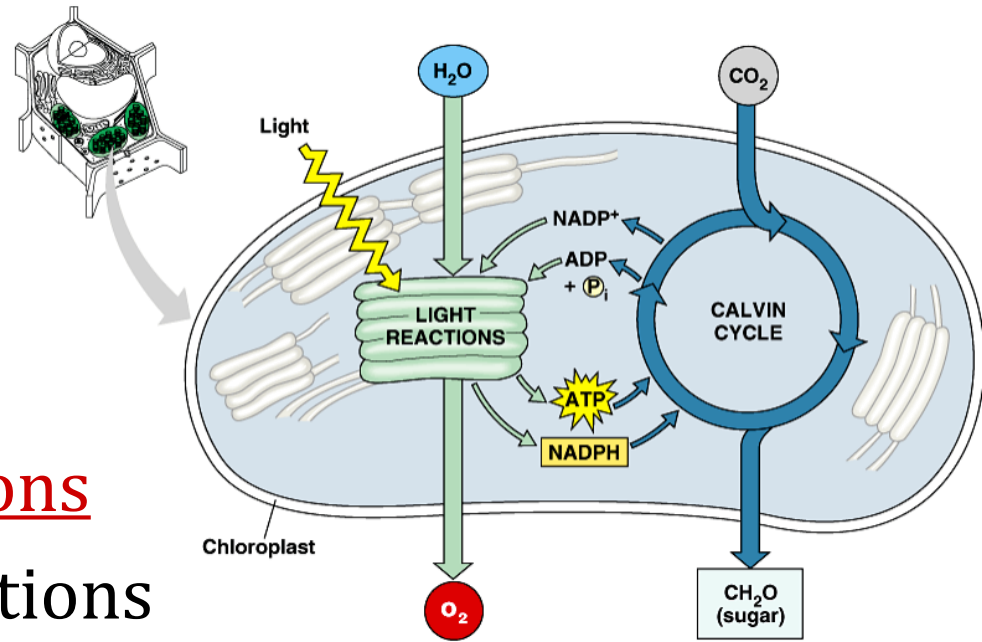


- **Thylakoid membrane contains**

- chlorophyll molecules
- electron transport chain
- ATP synthase
 - H^+ gradient built up within thylakoid sac



Photosynthesis



- Light reactions
 - light-dependent reactions
 - energy production reactions
 - convert solar energy to chemical energy
 - ATP & NADPH
- Calvin cycle
 - light-independent reactions
 - sugar production reactions
 - uses chemical energy (ATP & NADPH) to reduce CO_2 & synthesize $C_6H_{12}O_6$

Putting it all together

